

E.G.S. PILLAY ENGINEERING COLLEGE

(Autonomous)

Approved by AICTE, New Delhi| Affiliated to Anna University, Chennai

Accredited by NAAC with A++Grade| Accredited by NBA T1

(CSE, EEE, MECH, ECE, CIVIL, IT)

NAGAPATTINAM–611002



B.E. Biomedical Engineering

Third Year–Fifth Semester

COURSE CODE	COURSE NAME	CATEGORY	L	T	P	C	MAX.MARKS		
							CA	ES	TOTAL
THEORY COURSES									
2302BM501	Diagnostic and Therapeutic Equipment II	PCC	3	0	0	3	40	60	100
2302BM502	Fundamental of Rehabilitation Healthcare Analytics	PCC	3	0	0	3	40	60	100
2302BM503	Bio Signal Processing	PCC	3	1	0	4	40	60	100
	Professional Elective I	PEC	3	0	0	3	40	60	100
	Professional Elective II	PEC	3	0	0	3	40	60	100
	Open Elective I	OEC	3	0	0	3	40	60	100
PRACTICAL COURSES									
2302BM551	Biomechanics Laboratory	PCC	0	0	2	1	60	40	100
2302BM552	Biomedical Digital Signal Processing Laboratory	PCC	0	0	2	1	60	40	100
2304GE501	Professional development course-III	EEC	0	0	2	1	100	--	100
TOTAL			18	1	06	22	460	440	900

L–Lecture| T–Tutorial| P–Practical| C–Credit| CA–Continuous Assessment |ES–End Semester

COURSE CODE	COURSE NAME	CATE GORY	L	T	P	C	MAX.MARKS		
							CA	ES	TOTAL
PROFESSIONAL ELECTIVE – I									
2303BM001	Biomaterials	PEC	3	0	0	3	40	60	100
2303BM002	Foundation Skills in Integrated Product Development	PEC	3	0	0	3	40	60	100
2303BM003	Clinical Engineering	PEC	3	0	0	3	40	60	100
2303BM004	Biomechanics	PEC	3	0	0	3	40	60	100
2303BM005	Speech and Audio Signal Processing	PEC	3	0	0	3	40	60	100
2303BM006	Medical Optics	PEC	3	0	0	3	40	60	100

COURSE CODE	COURSE NAME	CATE GORY	L	T	P	C	MAX.MARKS		
							CA	ES	TOTAL
PROFESSIONAL ELECTIVE – II									
2303BM007	Artificial Organs and Implants	PEC	3	0	0	3	40	60	100
2303BM008	Medical Device Design	PEC	3	0	0	3	40	60	100
2303BM009	Hospital Planning and Management	PEC	3	0	0	3	40	60	100
2303BM010	Rehabilitation Engineering	PEC	3	0	0	3	40	60	100
2303BM011	Biometrics	PEC	3	0	0	3	40	60	100
2303BM012	Wearable Devices	PEC	3	0	0	3	40	60	100

COURSE CODE	COURSE NAME	CATE GORY	L	T	P	C	MAX.MARKS		
							CA	ES	TOTAL
OPEN ELECTIVE									
2303BM013	Occupational safety and health in public health emergencies	OE	3	0	0	3	40	60	100

2302BM501	DIAGNOSTIC AND THERAPEUTIC EQUIPMENT – II (For B.E-BME)	L	T	P	C
		3	0	0	3
PRE REQUEST:					
Diagnostic and therapeutic equipment-1					
Biomedical instrumentation					

Text Books:
1. John G. Webster, “Medical Instrumentation Application and Design”, 4th edition, Wiley India PvtLtd, New Delhi, 2015
2. Joseph J. Carr and John M. Brown, “Introduction to Biomedical Equipment Technology”, Pearson education, 2012.
References:
1. Leslie Cromwell, “Biomedical Instrumentation and measurement”, 2nd edition, Prentice hall of India, New Delhi, 2015.
2. Richard Aston “Principles of Biomedical Instrumentation and Measurement”, Merrill Publishing Company, 1990
3.. L.A Geddas and L.E.Baker “Principles of Applied Biomedical Instrumentation” 2004.
4.. Myer Kutz “Standard Handbook of Biomedical Engineering & Design”, McGraw-Hill Publisher, 2003.
5. Khandpur R.S, “Handbook of Biomedical Instrumentation”, 3rd edition, Tata McGraw-Hill, New Delhi, 2014.

2302BM502	FUNDAMENTAL OF REHABILITATION HEALTHCARE ANALYTICS (For B.E-BME)	L	T	P	C
		3	0	0	3
PRE REQUISITES:					
Human Anatomy and Physiology					
COURSE OBJECTIVES:					
	1. Understand the fundamentals of data mining techniques in healthcare.				
	2. Understand the healthcare data analytics approaches used				
	3. Analyze the effective information retrieval from healthcare data using data analytics approaches				
COURSE OUTCOME:					
On the successful completion of the course, students will be able to					

CO1	Understand the fundamental concepts of healthcare data analytics
CO2	Analyze the effectiveness of data mining in clinical and non-clinical applications
CO3	Categorize the approaches to retrieve data from social media.
CO4	Analyze the process of advanced healthcare data analytics
CO5	Analyze the information present based on visual characteristics of an expert system.

COs Vs POs MAPPING:

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1			1	1				1	1
CO2	3	2	1			1	1				1	1
CO3	3	2	1			1	1				1	1
CO4	3	2	1			1	1				1	1
CO5	3	2	1			1	1				1	1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	1	1	
CO2	1	1	
CO3	1	1	
CO4	1	1	
CO5	1	1	

COURSE CONTENTS:

Module I	INTRODUCTION TO HEALTH CARE DATA ANALYTICS	9 Hours
Introduction, Healthcare data sources and basic analytics, Advanced data analytics for healthcare, Applications and practical systems for healthcare, resources for healthcare data analytics-Healthcare data sources and Basic analytics: Electronic Health Records (EHR), Components of EHR, Coding systems, Benefits of EHR, Barriers to Adopting EHR, Challenges using EHR data.		
Module II	MINING OF SENSOR DATA AND DATA MINING	9 Hours
Introduction, Scopes and Challenges in mining sensor data, Challenges in Healthcare data analytics, sensor data mining applications, Nonclinical Health care applications, Introduction to data mining, Natural language processing, Mining information from clinical text and current methodologies, Informatics for integrating Biology, Challenges of processing clinical reports, Clinical applications.		
Module III	SOCIAL MEDIA ANALYTICS	9 Hours
Introduction, Social media analysis for detection and tracking of infectious disease outbreaks: Outbreak detection, Analyzing and tracking outbreaks, syndromic surveillance systems based on social media, Social medical analysis for public health research: Topic models for analyzing health-related contents, detecting reports of adverse medical events and drug reactions, characterizing life style and well-being, analysis of social media use in healthcare.		
Module IV	ADVANCED DATA ANALYTICS FOR HEALTH CARE	9 Hours
Introduction- Basic statistical prediction models: Linear regression, generalized additive model, Logistic regression: Multi class, polytomous and ordered Logistic regression, Bayesian models, Markov Random Fields, Alternative clinical prediction models: Decision trees, ANN, cost sensitive learning, Advanced prediction models, survival models, Evaluation and validation.		
Module V	VISUAL ANALYTICS AND INFORMATION RETRIEVAL FOR HEALTH CARE	9 Hours
Introduction, Medical data visualization, Visual analytics in Healthcare, Introduction to information retrieval, Knowledge based information in healthcare and biomedicine, content of knowledge- based information resources, indexing, retrieval: Exact match and partial match retrieval, Evaluation: system- oriented evaluation, user oriented evaluation, Introduction to clinical decision support system		
		Total:45Hours
FURTHER READING: Medical device design		

Text Books:
1. Chandan K Reddy and Charu C. Aggarwal, Healthcare Data Analytics, CRC Press, 2015
2. Vikas Kumar, Health care Analytics Made simple, Pack Publishing, 2018
3. Raymond A Gensinger, Analytics in Healthcare: An Introduction, HIMSS, 2014
References:
1. Tinglong Dai, Sridhar Tayur, Hand book of Healthcare Analytics, Wiley, 2018
2. Hui Yang, Eva K, Lee, Healthcare Analytics: From Data to Knowledge to Healthcare Improvement, Wiley, 2016.

2302BM503	BIO SIGNAL PROCESSING (For B.E-BME)	L	T	P	C
		3	1	0	4
PRE REQUISITE:					
Probability and statistics					
Signals and system					
COURSE OBJECTIVES:					
	1. To learn discrete Fourier transform, properties and its computation.				
	2. To know the characteristics of IIR filter and to learn the design of IIR filters for filtering undesired signals.				
	3.Understand the sources, characteristics and conversion process of biomedical signals.				
	4. Learn the basics of signal averaging and data compression techniques.				
COURSE OUTCOME:					
On the successful completion of the course, students will be able to					
CO1	Design the discrete Fourier transform, properties and its computation				
CO2	Design the different types of analog and digital IIR filters				
CO3	Design the FIR filter based on the given specifications				
CO4	Discuss about the Nature of Biomedical Signals				

C05	Discuss time domain and frequency domain filters to remove noise and artifacts from biomedical signals												
Cos VsPos MAPPING:													
Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
C01	3	3	3	3	2	-	-	-	2	-	1	2	
C02	3	3	2	2	-	-	-	-	2	-	1	2	
C03	3	3	3	3	3	-	-	-	2	-	1	2	
C04	3	3	3	-	-	-	-	-	2	-	1	2	
C05	3	3	2	-	-	-	-	-	2	-	1	2	
Cos Vs PSOs MAPPING:													
					COs	PSO1	PSO2	PSO3					
					C01	1	-	2					
					C02	1	-	2					
					C03	1	-	2					
					C04	1	-	2					
					C05	1	-	2					
COURSE CONTENTS:													
Module I	DISCRETE FOURIER TRANSFORM AND COMPUTATION											12 Hours	
Discrete Fourier Transform –properties, magnitude and phase representation – Computation of DFT using FFT algorithm – DIT &DIF using radix 2 FFT – Butterfly structure.													
Module II	INFINITE IMPULSE RESPONSE FILTER											12 Hours	
Analog Filter design – Butterworth filter – Chebyshev filter – Design of IIR filters from analog filters – Impulse Invariance – Bilinear Transformation – IIR realizations – Direct Form, Cascade Form and Parallel Form													
Module III	FINITE IMPULSE RESPONSE FILTER											12 Hours	
FIR Design – FIR design by windows – Rectangular window – Hamming window – Hanning window – Triangular window – Blackman window – Kaiser window – FIR design by frequency sampling method													
Module IV	INTRODUCTION TO BIOMEDICAL SIGNALS											12 Hours	
Nature of Biomedical Signals, Objectives and difficulties in Biomedical analysis. Electrocardiography: ECG leads systems, ECG signal characteristics. Signal Conversion: Simple signal conversion systems, Conversion requirements for biomedical signals, Signal conversion circuits.													
Module V	SIGNAL AVERAGING											12 Hours	
Basics of signal averaging, signal averaging as a digital filter, signal averaging as a typical average, software for signal averaging, limitations of signal averaging. Adaptive Noise Cancelling: Principle noise canceller model, 60-Hz adaptive cancelling using a sine wave model, other applications of adaptive filtering													
											Total:60 Hours		
FURTHER READING: Biomedical engineering, Medical diagnosis.													
Text Books:													
1. John G. Proakis & Dimitris G.Manolakis, —Digital Signal Processing – Principles, Algorithms & Applications, Fourth Edition, Pearson Education / Prentice Hall, 2007.													
2. Biomedical Digital Signal Processing- Willis I.Tompkins, PIIL 2011.													
3. Biomedical Signal Processing Principles and Techniques- D C Reddy, McGraw- Hill publications 2015.													
4.Biosignal Processing Principles and Practices (1 st Edition)Edited By Hualou Liang, Joseph D. Bronzino, Donald R. Peterson 2013													
Reference Book													
1. SalivahananS ,Vallavaraj A., Gnanapriya C, Digital Signal Processing, Tata McGraw- Hill, New Delhi, 2008.													
2. Biomedical Signal Analysis-Rangaraj M. Rangayyan, John Wiley & Sons 2022.													

3. Advanced Bio signal Processing- Amine NaitAli 2020.
4. 'Biomedical Signal Processing' A Modern Approach – Ganesh R.Naik and Wellington Pinheiro dos santos 2024
5. Bio signal and Medical image processing – Third Edition – John L Semmlow 2014

2302BM552	BIOMEDICAL SIGNAL PROCESSING LABORATORY (For B.E-BME)				L	T	P	C						
					0	0	2	1						
PRE REQUISITE:														
Probability and Stochastic Processes														
Biomedical Signals and system														
COURSE OBJECTIVES:														
	1.To learn discrete Fourier transform, properties and its computation.													
	2. To know the characteristics of IIR filter and to learn the design of IIR filters for filtering undesired signals													
COURSE OUTCOME:														
On the successful completion of the course, students will be able to														
CO1	Generation of various signals, Analysis of signal and system properties.													
CO2	Computation of circular and linear convolution.													
CO3	Determine the frequency transformation and Analysis of sampling rate.													
CO4	Design of digital filters.													
CO5	Analyze the power spectral density of the system.													
COs Vs POs MAPPING:														
	Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
	CO1	3	2	1	1	1	1	1		1	1	1	1	
	CO2	3	2	1	1	1	1	1		1	1	1	1	

CO3	3	2	1	1	1	1	1		1	1	1	1
CO4	3	2	1	1	1	1	1		1	1	1	1
CO5	3	2	1	1	1	1	1		1	1	1	1

COs Vs PSOs MAPPING:

	COs	PSO1	PSO2	PSO3
	CO1	1	1	2
	CO2	1	1	2
	CO3	1	1	2
	CO4	1	1	2
	CO5	1	1	2

COURSE CONTENTS:

LIST OF EXPERIMENTS:

1. Simulation and verify Generation of Signals
2. Simulate and verify , Properties of Discrete time Systems-Linearity, Stability, Causality &Time Variance.
3. Simulate and verify ,Sampling of an audio signal with different sampling rate and reconstruct the sampled signal.
4. Compute and verify DFT of a signal using basic equation and FFT & power spectrum estimation using DFT
5. Design and Simulation of IIR filters.
6. Design and Simulation of FIR filters.
7. Simulate and verify ,Multirate signal processing-Down sampling , Up sampling , Decimation and Interpolation
8. Simulate and verify ,Arithmetic operations in DSPs

Total :30 Hours

References:

1.J.G. Proakis and D.G. Manolakis, „Digital Signal Processing Principles, Algorithms and Applications“, Pearson Education, New Delhi, PHI. 2003.
2. S.K. Mitra, „Digital Signal Processing – A Computer Based Approach“, McGraw Hill Edu, 2013.

2302BM551		BIOMECHANICS LABORATORY (For B.E-BME)						L	T	P	C		
								0	0	2	1		
PRE REQUISITE:													
Human anatomy and physiology													
COURSE OBJECTIVES:													
		1.To study the mechanics involved with various physiological systems.											
		2.To gain knowledge in deriving the mathematical models related to blood vessels.											
COURSE OUTCOME:													
On the successful completion of the course, students will be able to													
CO1	Understand the basics of MATLAB for Mechanics												
CO2	Simulate and Understand the mechanics of bones												
CO3	Simulate and Understand the mechanics of Soft Tissues												
CO4	Simulate and Understand the mechanics of Joints												
CO5	Simulate FEA models												
COs Vs POs MAPPING:													
	Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
	CO1	3	2	1	1	1	1	1				1	1
	CO2	3	2	1	1	1	1	1				1	1
	CO3	3	2	1	1	1	1	1				1	1
	CO4	3	2	1	1	1	1	1				1	1

COURSE CODE	COURSE NAME	CATE GORY	L	T	P	C	MAX.MARKS		
							CA	ES	TOTAL
PROFESSIONAL ELECTIVE – I									
2303BM001	Biomaterials	PEC	3	0	0	3	40	60	100
2303BM002	Foundation Skills in Integrated Product Development	PEC	3	0	0	3	40	60	100
2303BM003	Clinical Engineering	PEC	3	0	0	3	40	60	100
2303BM004	Biomechanics	PEC	3	0	0	3	40	60	100
2303BM005	Speech and Audio Signal Processing	PEC	3	0	0	3	40	60	100
2303BM006	Medical Optics	PEC	3	0	0	3	40	60	100

2303BM001	BIOMATERIALS (For B.E-BME)	L	T	P	C
		3	0	0	3
PRE REQUISITE:					
	Human Anatomy and Physiology				
	Biosciences in Medical Engineering				

COURSE OBJECTIVES:													
	1.To learn about different types of Biomaterials												
	1.To Understand different metals, ceramics ,Polymer and its characteristics as biomaterials												
	3.To understand the Biomedical application of different textile structures												
COURSE OUTCOME:													
On the successful completion of the course, students will be able to													
CO1	Understand the structure of bio-materials and its bio-compatibility.												
CO2	Apply appropriate implant materials in medical field based on their properties												
CO3	Analyze polymeric materials and its combinations that could be used as a tissue replacement .												
CO4	Analyse soft and hard tissue replacement implants in biomedical applications.												
CO5	Apply material properties of biopolymers suitable for implantable, non, implantable and drug delivery textiles												
COs Vs POs MAPPING:													
	Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
	CO1	3	2	1			1	1				1	1
	CO2	3	2	1			1	1				1	1
	CO3	3	2	1			1	1				1	1
	CO4	3	2	1			1	1				1	1
	CO5	3	2	1			1	1				1	1
COs Vs PSOs MAPPING:													
					COs	PSO1	PSO2	PSO3					
					CO1	1		1					
					CO2	1		1					
					CO3	1		1					
					CO4	1		1					
					CO5	1		1					
COURSE CONTENTS:													
Module I	STRUCTURE OF BIO-MATERIALS AND BIO-COMPATIBILITY											9 Hours	
Definition and classification of bio-materials, mechanical properties, viscoelasticity, wound healing process, body response to implants, blood compatibility, HLA compatibility.													
Module II	IMPLANT MATERIALS											9 Hours	
Metallic implant materials, stainless steels, Ti-based alloys, ceramic implant materials, aluminium oxides, hydroxyapatite, glass ceramics, carbons. Polymerization, polyamides, Acrylic polymers, Hydrogels, rubbers, high strength, thermoplastics, medical applications.													
Module III	BIOPOLYMERS											9 Hours	
Biopolymers: classification and their properties, requirements, and applications, testing methods; In vitro tests , direct contact, agar diffusion & elution methods , in vivo assessment of tissue compatibility. collagen and elastin. Materials for ophthalmology: contact lens, Intra ocular lens. Membranes for plasma separation and blood oxygenation													
Module IV	TISSUE REPLACEMENT IMPLANTS											9 Hours	
Small intestinal submucosa and other decellularized matrix biomaterials for tissue repair. Soft tissue replacements, types of transplant by stem cell, sutures, surgical tapes, Tissue adhesive/glue. Percutaneous and skin implants, maxillofacial augmentation, Vascular grafts, hard tissue replacement Implants, point replacements.													
Module V	IMPLANTABLES, NON, IMPLANTABLES AND DRUG DELIVERY											9 Hours	

Bandages, types, properties and applications; compression garments, types, properties and applications; sutures: types and properties; implantable textiles: hernia mesh , vascular prostheses , stents; Extra corporeal materials: Cartilage nerves , liver ligaments, kidney, tendons, cornea; Drug delivery textiles: classification , mechanism various fabrication methods , characterization , applications.
Total: 45 Hours
FURTHER READINGS: Nano materials for medical device design
Text books:
1.Sujata V. Bhatt, Biomaterials, Narosa Publishing House,7 th Edition,2005
2.Guigen Zhang, Michael J. Yaszemski, Biomaterial: An Introduction to Biomaterial in Medicine, Elsevier, 2020
References:
1.Park Joseph D.Bronzino, Biomaterials-Principles and Applications, CRC press,2003.
2. John Enderle, Joseph D. Bronzino, Susan M.Blanchard, —Introduction to Biomedical Engineeringll, Elsevier, 2005.
3. Sreeram Ramakrishna, Murugan Ramalingam, T. S. Sampath Kumar, and Winston O. Soboyejo, Biomaterials: A Nano Approachll, CRC Press, 2010
4.Joon B. Park., and Joseph D. Bronzino., Biomaterials , Principles and Applications , CRC Press, Boca Raton London, New York, Washington, D.C. 2002.

2303BM002	FOUNDATION SKILLS IN INTEGRATED PRODUCT DEVELOPMENT (For B.E-BME)	L	T	P	C
		3	0	0	3
PRE REQUISITE:					
Human physiology and anatomy					
Biomechanics					
COURSE OBJECTIVES:					
	1. To understand the global trends of product development and methodologies of various types of new product.				
	2. To understand requirement engineering and know how to collect, analyze and arrive at requirements for new product development and				

	3. To develop documentation, test specifications and coordinate with various teams to validate and sustain up to the EoL support activities for engineering customer												
COURSE OUTCOMES:													
On the successful completion of the course the students will be able to,													
CO 1	Define, formulate and analyze a problem												
CO 2	Solve specific problems independently or as part of a team												
CO 3	Gain knowledge of the Innovation & Product Development process in the Business Context												
CO 4	Work independently as well as in teams												
CO 5	Manage a project from start to finish												
COs Vs POs MAPPING:													
Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
CO1	3	2	1			1	1	1			1	1	
CO2	3	2	1			1	1	1			1	1	
CO3	3	2	1			1	1	1			1	1	
CO4	3	2	1			1	1	1			1	1	
CO5	3	2	1			1	1	1			1	1	
COs Vs PSOs MAPPING													
	COs	PSO1	PSO2	PSO3									
	CO1		1	1									
	CO2		1	1									
	CO3		1	1									
	CO4		1	1									
	CO5		1	1									
COURSE CONTENTS:													
Module I	FUNDAMENTALS OF PRODUCT DEVELOPMENT											9 Hours	
Global Trends Analysis and Product decision – Social Trends – Technical Trends-Economic Trends – Environmental Trends – Political/Policy Trends – Introduction to Product Development Methodologies and Management – Overview of Products and Services – Types of Product Development – Overview of Product Development methodologies – Product Life Cycle – Product Development Planning and Management.													
Module II	REQUIREMENTS AND SYSTEM DESIGN											9 Hours	
Requirement Engineering - Types of Requirements - Requirement Engineering - traceability Matrix and Analysis - Requirement Management - System Design & Modelling - Introduction to System Modelling - System Optimization - System Specification - Sub-System Design - Interface Design.													
Module III	DESIGN AND TESTING											9 Hours	
Conceptualization – Industrial Design and User Interface Design – Introduction to Concept generation Techniques – Challenges in Integration of Engineering Disciplines – Concept Screening and Evaluation – Detailed Design – Component Design and Verification -Mechanical, Electronics and Software Subsystems – High Level Design/Low Level Design of S/W Program – Types of Prototypes, S/W Testing- Hardware Schematic, Component design, Layout and Hardware Testing – Prototyping – Introduction to Rapid Prototyping and Rapid Manufacturing – System Integration, Testing, Certification and Documentation													
Module IV	SUSTENANCE ENGINEERING AND END-OF-LIFE (EOL) SUPPORT											9 Hours	
Introduction to Product verification processes and stages - Introduction to Product Validation processes and stages - Product Testing Standards and Certification - Product Documentation - Sustenance - Maintenance and Repair – Enhancements - Product EoL - Obsolescence Management – Configuration Management - EoL Disposal													
Module V	BUSINESS DYNAMICS – ENGINEERING SERVICES INDUSTRY											9 Hours	
The Industry – Engineering Services Industry – Product Development in Industry versus Academia - The IPD Essentials – Introduction to Vertical Specific Product Development processes - Manufacturing/Purchase and Assembly of Systems – Integration of Mechanical, Embedded and Software Systems – Product Development Trade-offs – Intellectual Property Rights and Confidentiality – Security and Configuration Management.													
Total:45Hours													

FURTHER READINGS: Medical device design
Text Books:
1. Book specially prepared by NASSCOM as per the MoU.
2. Karl T Ulrich and Stephen D Eppinger, “Product Design and Development”, Tata McGraw Hill, Fifth Edition, 2011.
3. John W Newstorm and Keith Davis, “Organizational Behavior”, Tata McGraw Hill, Eleventh Edition, 2005
References:
1.Hiriyappa B, “Corporate Strategy – Managing the Business”, Author House, 2013.
2.Peter F Drucker, “People and Performance”, Butterworth – Heinemann [Elsevier], Oxford, 2004.
3.Vinod Kumar Garg and Venkita Krishnan N K, “Enterprise Resource Planning -Concepts”, Second Edition, Prentice Hall, 2003.
4.Mark S Sanders and Ernest J McCormick, “Human Factors in Engineering and Design”, McGraw Hill Education, Seventh Edition, 2013.

2303BM003	CLINICAL ENGINEERING (For B.E-BME)	L	T	P	C
		3	0	0	3
PRE REQUISITE:					
Human physiology and anatomy					
Biomechanics					
Biochemistry					
Hospital training					
COURSE OBJECTIVES:					
	1.Understanding of the clinical engineering profession, qualifications, roles, activities, and expectations				
	2.Enhance the students to practice medical equipment work as a team to address problems and errors in medical devices				
	3.Design better medical devices and explore the Health Technology Management systems				
COURSE OUTCOMES:					
On the successful completion of the course the students will be able to,					
CO1	State the role of clinical engineers and discuss the basic concepts of medical and healthcare technology				
CO2	Give the program and framework to recognize the errors of medical equipment.				
CO3	State the issues or errors in patient safety and formulate patient safety package system				
CO4	Define the problem precisely and examine the possible issues using program indicators				

2303BM004	BIOMECHANICS (For B.E, BME)								L	T	P	C
									3	0	0	3
PRE REQUISITE:												
Solid state physics												
Human physiology and anatomy												
COURSE OBJECTIVES:												
	1.Understand the basic principle of mechanics needed for biomedical engineers.											
	2.Discuss the bio-fluid mechanics of human anatomy, the mechanics of tissues, muscles and bones.											
	3. Illustrate the Skeletal joints of human and Design of a Computer work station											
COURSE OUTCOMES:												
On the successful completion of the course the students will be able to,												
CO1	Understand the principles of mechanics.											
CO2	Outline the principles of bio fluid dynamics.											
CO3	Explain the fundamentals of bio-solid mechanics.											
CO4	Interpret the knowledge of joint mechanics.											
CO5	Give Examples of computational mathematical modeling applied in biomechanics.											
COs Vs POs MAPPING:												
Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1		1	1				1	1
CO2	3	2	1	1		1	1				1	1
CO3	3	2	1	1		1	1				1	1
CO4	3	3	1	1		1	1				1	1

CO5	3	2	1	1		1	1				1	1	
COs Vs PSOs MAPPING:													
					COs	PSO1	PSO2	PSO3					
					CO1		1	1					
					CO2		1	1					
					CO3		1	1					
					CO4		1	1					
					CO5		1	1					
COURSE CONTENTS:													
Module I		INTRODUCTION TO MECHANICS									9 Hours		
Bio-fluid Mechanics Newton's Law; Stress, Strain, Elasticity; Hooke's Law; Viscosity; Newtonian Fluid; Non- Newtonian Fluid; Viscoelastic Fluids; Velocity and Pressure of Blood Flow; Resistance Against Flow.													
Module II		BIOFLUID MECHANICS									9 Hours		
Cardiovascular and Respiratory Mechanics Mechanical Properties of Blood Vessels: Arteries, Arterioles, Capillaries and Veins; Function of Cardiac Chambers & Valves; Mechanics of Angiography and Angioplasty; Stent Deployment & Prosthetic Replacement of Cardiac Valves; Alveoli Mechanics; Interaction of Blood and Lung; Breathing Mechanism; Airway Resistance; Working Dynamics of Spirometer; Ventilators													
Module III		BIOSOLID MECHANICS									9 Hours		
Soft Tissue Mechanics Pseudoelasticity; Non-Linear Stress-Strain Relationship; Visco elasticity; Structure Function and Mechanical Properties of Skin; Ligaments and Tendons.													
Module IV		BIOMECHANICS OF JOINTS									9 Hours		
Orthopedic Mechanics Mechanical Properties of Cartilage; Stress-Strain Analysis; Mechanical Properties of Bones and Implants; Design Consideration of Stress Shielding; Kinetics and Kinematics of Joints; Lubrication of Joints; Foot Mechanics.													
Module V		MODELING AND ERGONOMICS									9 Hours		
Introduction to Finite Element Analysis, finite element analysis of lumbar spine ; Ergonomics–Musculoskeletal disorders, Ergonomic principles contributing to good work place design.													
Total:45Hours													
FURTHER READINGS:													
Basic orthopedic biomechanics & mechano-biology, Mechanical tribology: materials, characterization, and applications													
Text Books:													
		1. Y.C.Fung,—Bio-Mechanics-Mechanical Properties of Tissues, Springer-Verlag, 1998.											
		2. Subrata Pal,—Textbook of Biomechanics, Viva Books Private Limited, 2009.											
		3. Hall, S.J. and Lysell, D., 1995. Basic biomechanics (Vol. 2). St. Louis: Mosby											
		4. . Knudson, D., 2007. Fundamentals of biomechanics. Springer Science & Business Media.											
References:													
		1. Krishna B.Chandran, Ajit P.Yoganathan and Stanley E. Rittgers Bio-fluid Mechanics: The Human Circulation, Taylor and Francis, 2007.											
		2. Sheraz S. Malik and Shahbaz S. Malik, —Orthopaedic Biomechanics Made Easy, Cambridge University Press, 2015											
		3. Jay D.Humphrey, Sherry De Lange, An Introduction to Biomechanics: Solids and Fluids, Analysis and Design, Springer Science Business Media, 2004.											

2303BM005	SPEECH & AUDIO SIGNAL PROCESSING (For B.E, BME)									L	T	P	C
										3	0	0	3
PRE REQUISITE:													
Human physiology and anatomy													
Signals and systems													
Bio signal processing													
COURSE OBJECTIVES:													
	1.Understand basics of Anatomy & physiology of speech production and audio perception mechanism												
	2.Implement different methods of processing speech and audio signals using time & transform domain techniques												
	3.Discuss the methods of parametric representation of speech signals and speech coding												
COURSE OUTCOMES:													
On the successful completion of the course the students will be able to,													
CO1	Familiarize the basic speech production and audio perception mechanism												
CO2	Implement different methods of processing speech and audio signals using time & transform domain techniques												
CO3	Analyse the methods of parametric representation of speech signals												
CO4	Apply the concepts of speech coding												
CO5	Apply various speech and audio signal processing techniques in biomedical applications												
COs Vs POs MAPPING:													
Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
CO1	3	2	1			1	1				1	1	
CO2	3	2	1			1	1				1	1	
CO3	3	2	1			1	1				1	1	
CO4	3	3	1			1	1				1	1	
CO5	3	2	1			1	1				1	1	
COs Vs PSOs MAPPING:													
					CO	PSO1	PSO2	PSO3					

	s				
	CO1	1	1		
	CO2	1	1		
	CO3	1	1		
	CO4	1	1		
	CO5	1	1		
COURSE CONTENTS:					
Module I	INTRODUCTION TO SPEECH PRODUCTION				9 Hours
Anatomy & physiology of speech production. Speech Fundamentals: Articulatory Phonetics – Production and Classification of Speech Sounds; Articulation, Voicing, Articulatory model. Acoustic Phonetics – acoustics of speech production; Acoustic theory of speech production-Excitation, Vocal tract model, Formant structure, Pitch.					
Module II	AUDIO PERCEPTION				9 Hours
Basic anatomy of hearing System. Psycho-acoustic analysis: Sound pressure level & loudness, Auditory Filter Banks, Critical Band Structure, Absolute Threshold of Hearing, Simultaneous Masking, Temporal Masking. Speech perception: Vowel Perception					
Module III	TIME AND FREQUENCY DOMAIN METHODS FOR SPEECH AND AUDIO PROCESSING SPEECH ANALYSIS				9 Hours
Short-Time Speech Analysis, Time domain analysis- Short time energy, short time average magnitude, short time zero crossing rate, short time ACF. Frequency domain analysis- Filter Banks, STFT, Spectrogram, Formant Estimation & Analysis, Cepstral Analysis-MFCC					
Module IV	PARAMETRIC REPRESENTATION OF SPEECH				9 Hours
AR Model, ARMA model. LPC Analysis- LPC model, Auto correlation method, Covariance method. Levinson-Durbin Algorithm, Sinusoidal Model, HMM. Applications of LPC parameters as pitch detection and formant analysis					
Module V	SPEECH CODING				9 Hours
Phase Vocoder, LPC, Sub-band coding, Adaptive Transform Coding, Harmonic Coding, Vector Quantization based Coders, CELP. Biomedical applications of speech signal processing: - Detection of neurodegenerative diseases, emotion recognition, Speaker Verification.					
				Total:	45Hours
FURTHER READINGS: Speech processing application in medical Application					
Text Books:					
1.Thomas F. Quatieri, Discrete-Time Speech Signal Processing: Principles and Practice, Prentice Hall; ISBN: 013242942X; 1st Edition, 2001					
2. Rabiner and Schafer, Digital Processing of Speech Signals, Prentice Hall, 2 nd Edition					
3. Deller J. R. Proakis J. G. and Hanson J. H, Discrete Time Processing of Speech Signals, Wiley Inter science, 2010					
4. Nelson Morgan and Ben Gold, Speech and Audio Signal Processing: Processing and Perception of Speech and Music, John Wiley & Sons, ISBN: 0471351547, 2nd Edition, 2011					
References:					
1.Douglas O'Shaughnessy, Speech Communications: Human & Machine, IEEE Press, 2nd Edition, 1999; ISBN: 0780334493.					
2.Donald G. Childers, Speech Processing and Synthesis Toolboxes, John Wiley & Sons, September 1999; ISBN: 0471349593					
3.Rabiner and Juang, Fundamentals of Speech Recognition, Prentice Hall, 1st Edition, 1994 ISBN-10 – 0130151572					

Module II	INSTRUMENTATION IN PHOTONICS	9 Hours
Instrumentation for absorption, scattering and emission measurements, excitation light sources–high pressure arc lamp, solid state LEDs, Lasers, optical filters, polarizer, solidstate detectors, time resolved and phase resolved detectors.		
Module III	APPLICATIONS OF LASERS	9 Hours
Lasers in ophthalmology, Dermatology, Dentistry, Urology, Otolaryngology, Tissue welding and Soldering.		
Module IV	OPTICAL TOMOGRAPHY	9 Hours
Optical coherence tomography, Elastography, Doppler optical coherence tomography, Application towards clinical imaging.		
Module V	SPECIAL OPTICAL TECHNIQUES	9 Hours
Near field imaging of biological structures, invitro clinical diagnostic, fluorescent spectroscopy, photodynamic therapy.		
		Total:45 Hours
FURTHER READINGS: Medical image analysis		
Text Books:		
1.Tuan VoDirh,—Biomedical Photonics—Hand book,CRC Press,Bocaraton,2014.		
2. Paras N.Prasad,—Introduction to Bio photonics,A.John Wiley and Sons,Inc.Publications,2003		
References:		
1. Markolf H.Niemz,—Laser-Tissue Interaction Fundamentals and Applications, Springer,2007		
2.G.David Baxter—Therapeutic Lasers Theory and practice, Churchill Livingstone publications Edition- 2001.		
3. Leon Goldman, M.D.,& R.James Rockwell,Jr.,—Lasers in Medicine, Gordon and Breach, Science Publishers Inc., 1975.		

COURSE CODE	COURSE NAME	CATE GORY	L	T	P	C	MAX.MARKS		
							CA	ES	TOTAL
PROFESSIONAL ELECTIVE – II									
2303BM007	Artificial Organs and Implants	PEC	3	0	0	3	40	60	100
2303BM008	Medical Device Design	PEC	3	0	0	3	40	60	100
2303BM009	Hospital Planning and Management	PEC	3	0	0	3	40	60	100
2303BM0010	Rehabilitation Engineering	PEC	3	0	0	3	40	60	100
2303BM0011	Biometrics	PEC	3	0	0	3	40	60	100
2303BM0012	Wearable Devices	PEC	3	0	0	3	40	60	100

2303BM007	ARTIFICIAL ORGANS AND IMPLANTS (For B.E.,BME)										L 3	T 0	P 0	C 3
PRE REQUISITE:														
1.Rehabilitation engineering														
COURSE OBJECTIVES:														
	1.To have an overview of artificial organs and transplants													
	2.To describe the principles of implant design and various organs replacement concept													
	3.To study about physical parameters for concept design of artificial organs and the concept of biocompatibility													
COURSE OUTCOMES:														
On the successful completion of the course the students will be able to,														
CO1	Learner will be able to understand the principles and biology underlying the design of implants and artificial organs.													
CO2	Able to understand the principles of implant design with a case study													
CO3	Will be familiar about various organs replacement concept													
CO4	Knowledge about physical parameters for concept design of artificial organs													
CO5	Able to understand the concept of biocompatibility and the methods of biomaterial testing													
COs Vs POs MAPPING:														
	Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
	CO1	3	2	1					1			1	1	
	CO2	3	2	1					1			1	1	
	CO3	3	2	1					1			1	1	
	CO4	3	3	1					1			1	1	
	CO5	3	2	1					1			1	1	
COs Vs PSOs MAPPING:														
					COs	PSO1	PSO2	PSO3						
					CO1	1	1							
					CO2	1	1							
					CO3	1	1							
					CO4	1	1							
					CO5	1	1							
COURSE CONTENTS:														
Module I	ARTIFICIAL HEART & CIRCULATORY ASSIST DEVICES											9 Hours		
Engineering Design of artificial Heart & Circulatory Assist Devices; Detailed Design to execute the plant; Heart Assist Technology; Blood Pumps; Prosthetic Heart Valves.														
Module II	ARTIFICIAL BLOOD & COCHLEAR IMPLANT											9 Hours		
Blood components & characteristics; Oxygen carrying plasma expanders; Blood substitutes; Crystalloid & colloidal solutions as volume expanders; Artificial oxygen carriers; Fluorocarbons ; Hemoglobin based artificial blood. Cochlear Implant: Introduction; candidates for implant; the auditory system; the auditory periphery; theory of operation; evaluation of cochlear prosthesis; benefits & risks of implantation; the cost of implantation; the future of cochlear prosthesis.														
Module III	ARTIFICIAL SKIN											9 Hours		
Structure & functions of skin; Characteristics & clinical use of skin substitutes; Two conceptual stages in the treatment of massive skin loss; Skin substitutes: characteristics & uses, types of skin substitutes.														

Module IV	ARTIFICIAL PANCREAS & ARTIFICIAL LUNGS	9 Hours
Parenteral Structure & function of Pancreas; Endocrine pancreas & insulin secretion; Diabetes; Insulin therapy; Insulin administration systems; Insulin production systems. Artificial Lungs: Gas exchange systems; Cardiopulmonary Bypass; Oxygen & CO2 transport; Coupling of oxygen & CO2 exchange, Shear-Induced Transport Augmentation and Devices for Improved Gas Transport.		
Module V	BIOMATERIAL TESTING AND ARTIFICIAL ORGANS IMPLANTS	9 Hours
Testing of biomaterials: In-vitro, in-vivo preclinical tests - biocompatibility - methods for improvement, surface modification of materials - implant retrieval and evaluation. Artificial Heart, eye and ear implants, artificial pancreas, ophthalmic implantation, dental implantation, insulin administration devices, extracorporeal artificial organs, neural prostheses.		
		Total: 45 Hours
FURTHER READINGS: Rehabilitation Engineering		
Text Books:		
1.Bronzino, J. D. (2000). The Biomedical Engineering Handbook. Germany: CRC Press.		
References:		
1. Miller, G. E. (2006). Artificial Organs. United States: Morgan & Claypool Publishers.		
2.Schoen, F. J., Ratner, B. D., Hoffman, A. S., Lemons, J. E. (2004).		
3.Biomaterials Science: An Introduction to Materials in Medicine. Netherlands: Elsevier		
4.Artificial Organs. (2009). Netherlands: Springer London.		

2303BM008	MEDICAL DEVICE DESIGN (For B.E, BME)	L	T	P	C
		3	0	0	3
PRE REQUISITE:					
Biomedical instrumentation					
Biosensors and measurements					

COURSE OBJECTIVES:

	1.To know about the Medical product design and development
	2.Patient safety and regulatory aspects followed in hospitals
	3.Professional ethics to be followed by Biomedical Engineers

COURSE OUTCOMES:

On the successful completion of the course the students will be able to,

CO1	Implement Good Design Practice in medical product design
CO2	Apply appropriate methodologies for Product development
CO3	Analyse important regulatory schemes to be followed in medical device design
CO4	Apply testing, validation and market analysis for developed product
CO5	Analyse challenges in converting innovation into product in Healthcare sector

COs Vs POs MAPPING:

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	1						1	1
CO2	3	2	1	1	1						1	1
CO3	3	2	1	1	1						1	1
CO4	3	2	1	1	1						1	1
CO5	3	2	1	1	1						1	1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	1	1	
CO2	1	1	
CO3	1	1	
CO4	1	1	
CO5	1	1	

COURSE CONTENTS:

Module I	INTRODUCTION TO MEDICAL PRODUCT DESIGN	9 Hours
Design and Product Life Cycle, Design Process, Understanding the innovation cycle, Good Design Practice. Understanding, Screening Needs, Technical Requirements, Concept Generation, Innovation Survey Questionnaire, Morphological Matrix, QFD, Concept Analysis and validation, Concept Modeling, Concept Screening & Validation.		
Module II	PRODUCT DEVELOPMENT	9 Hours
Breakthrough Products, Platform Products, Front End of Innovations, Fuzzy Front End, Generic Product Development Process, Variants of Development Processes, Good Documentation Practice, and Prototyping Specifications, Prototyping, Medical Device standards, Quality management systems(ISO 13485), Medical Device Classification, Design of Clinical Trials.		
Module III	REGULATORY SCHEMES	9 Hours
DesignControl&RegulatoryRequirements,DocumentationinMedicalDevices,Regulatorypathways, Biomedical Evaluation of Medical Devices, ISO Medical Devices, Applications of Risk Management to Medical Devices (ISO 14971), Electrical Safety Standard, IEC60601-1, IEC60601-2, IEC60601-6, Protection of Electrical and Electronic Parts, Assemblies and Equipments (ESD S20.20-2014).		
Module IV	SCALABLE PRODUCT DEVELOPMENT	9 Hours
Design for manufacturing, Design for assembly, Design for Serviceability, Design for usability, Medical Device Verification & Validation, Product Testing & Regulatory		

compliance, Clinical trial & validation, Device Certification.		
Module V	PRACTICAL CHALLENGES ON MEDICAL DEVICE DEVELOPMENT	9 Hours
Product life cycle, Challenges in Practicing International Regulatory Requirements, Risk Management: Integration of Risk Management into the supporting QMS, Use of Codes to Identify Medical Devices, Application of Risk Management throughout product life cycle.		
		Total:45 Hours
FURTHER READINGS: Module design for medical diagnostics		
References:		
1.JohnG.Webster,Medical Instrumentation: Application and Design,5 th Edition, June 2020		
2.Peter J. Ogrodnik, Medical Device Design: Innovation from Concept to Market, Academic Press is an imprint of Elsevier, 1st edition 2013		
3.PaulH.King,RichardC.Fries,Arthur T.Johnson, Design of Biomedical Devices and Systems, CRC Press, Tailor and Francis Group, 3rd Edition, 2015		

2303BM009	HOSPITAL PLANNING AND MANAGEMENT (B.E-BME)	L	T	P	C
		3	0	0	3
PRE REQUISITE:					
Human physiology and anatomy					
Clinical engineering					
Hospital training					
COURSE OBJECTIVES:					
	1.To understand the fundamentals of hospital administration and management.				
	2.To explore various information management systems and relative supportive services.				
	3.To learn the quality and safety aspects in hospitals.				

1.R.C.Goyal, Hospital Administration and Human Resource Management, PHI-Fourth Edition, 2006.
2. G.D.Kunders, Hospitals Facilities Planning and Management- TMH, New Delhi- Fifth Reprint 2007.
3. Ramani KV, Hospital Management- Text and Cases, Pearson education, New Delhi, 2012.
4. Malhotra A K, Hospital Management- An Evaluation, Global India Publications, New Delhi, 2009.
5. Norman Metzger, Handbook of Health Care Human Resources Management, 2nd edition Aspen Publication Inc. Rockville, Maryland, USA, 1990.
6. Blane, David, Brunner, Health and social organization: Towards a Health Policy for the 21st Century, Eric Calrendon Press 2002

2303BM010	REHABILITATION ENGINEERING (B.E-BME)	L	T	P	C
		3	0	0	3
PRE REQUISITE:					

Human physiology and anatomy												
Biomechanics												
Clinical engineering												
Hospital training												
COURSE OBJECTIVES:												
	1.The course is designed to provide a brief and basic knowledge to understand musculoskeletal, neuromuscular, sensory disorders, prosthetics and orthotics and their applications.											
	2.To know the basic concept so that the students can implement their knowledge for higher studies in developing innovative and effective rehabilitation and assistive technologies											
	3.To DiscussVisual augmentation and advanced applications in rehabilitation engineering											
COURSE OUTCOMES:												
On the successful completion of the course the students will be able to,												
CO1	Study basics of Rehabilitation Engineering											
CO2	Learn the Orthotics & Ortho prosthetics											
CO3	Gain knowledge of the Wheeled Mobility											
CO4	Understand sensory augmentation and substitution											
CO5	Study recent developments in the field of rehabilitation engineering											
COs Vs POs MAPPING:												
Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1										1	1
CO2	1	2	2								1	1
CO3	1	2	2	1							1	1
CO4	1	2	2	1							1	1
CO5	1										1	1
COs Vs PSOs MAPPING:												
				COs	PSO1	PSO2	PSO3					
				CO1	1							
				CO2	1							
				CO3	1							
				CO4	1							
				CO5	1							
COURSE CONTENTS:												
Module I	INTRODUCTION TO REHABILITATION											9 Hours
Introduction Concepts and principles of rehabilitation engineering, Ergonomics - Positioning anatomical site, simplicity and intuitive operation, adaptability and flexibility, mental and chronological age appropriateness. Knowledge of disability act 1995 for physically disabled, visually impaired, hearing impaired and others. Rehabilitation Team.												
Module II	ORTHOTICS & ORTHOPROSTHETICS											9 Hours
Orthopedic prosthetics and orthotics in Rehabilitation Fundamentals, Applications: Computer Aides Engineering in customized component design. Intelligent prosthetic knee and hand. A self-aligning orthotic knee joint. FES System: Restoration of hand function; restoration of standing and walking. Hybrid Assistive systems (HAS) Active prostheses. Active Above knee Prosthesis. Myoelectric hand and arm prostheses. Orthotics - FO, AFO, TLSO, LSO												
Module III	WHEELED MOBILITY											9 Hours

History and Categories of Wheelchairs. Wheelchair Structure and Component Design. Ergonomics of wheel chair propulsion. Power Wheelchair Electrical System. Personal transportation. Tricycles.		
Module IV	SENSORY AUGMENTATION AND SUBSTITUTION	9 Hours
Visual System: Visual augmentation. Tactual vision substitution. Auditory vision substitution: Auditory System: Auditory augmentation. Cochlear implantation. Visual auditory substitution. Tactual auditory substitution. Tactual system: Tactual augmentation. Tactual substitution. Alternative and Augmentative communication, User Interface: Outputs: Acceleration Techniques.		
Module V	ADVANCED APPLICATIONS IN REHABILITATION ENGINEERING	9 Hours
Interfaces in Compensation for visual perception. Improvement of orientation and mobility. Computer - assisted lip reading. Brain - computer interface. Electronic Travel Applications (ETA) : Path Sounder, Laser Cane, Ultrasonic Torch, Sonic Guide, Light Probes, Nottingham Obstacle Sensor, Electro-cortical Prosthesis,. Polarized Ultrasonic Travel Aid.		
TOTAL:		45 Hours
FURTHER READINGS: Advanced equipment design		
Text & References:		
1. Bronzino, Biomedical Engineering, Hand Book IEEE Press Volume 1.		
2. Robinson C. J, Rehabilitation Engineering, CRC Press. 1995.		
3. Ballabio E. et al, Rehabilitation Technology, IOS Press. 1993.		
4. Handbook of Physical Medicine & Rehabilitation, W.B.Saunders Publication, 2003.		
5. Hanfred clynes, Biomedical Engineering System, McGraw Hill, 1999		

2303BM011	BIOMETRICS (For B.E.,BME)	L	T	P	C
		3	0	0	3
PRE REQUISITE:					
Image processing					
COURSE OBJECTIVES:					
1. To understand the general principles of design of biometric systems and the underlying trade-offs.					
2. To understand the technologies of fingerprint identification technology					

	3. To explain face recognition representations and determination
	4. To Discuss speech recognition and evaluations.
	5. To recognize multi biometrics and design aspects.

COURSE OUTCOMES:

On the successful completion of the course the students will be able to,

CO1	Explain fundamental concept of biometrics, its benefits and performance measures
CO2	Analyse fingerprint patterns, features and perform fingerprint matching
CO3	Demonstrate the working of face recognition technologies, classification and evaluation
CO4	Apply principles of voice scan, speaker recognition and access system performance.
CO5	Evaluate multi biometric fusion techniques and design integrated biometric system

COs Vs POs MAPPING:

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-		1	1	2		2	1	2
CO2	3	3	2	2	2		1			1	1	2
CO3	3	3	2	2	3		1	1		2	1	2
CO4	3	3	2	2	3		1		1	2	1	2
CO5	3	3	3	2	3	1	1	2	2	3	1	3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	3	2	2
CO2	3	3	2
CO3	2	3	3
CO4	2	3	3
CO5	3	3	3

COURSE CONTENTS:

Module I	INTRODUCTION TO BIOMETRICS	9 Hours
Introduction and back ground – biometric technologies – passive biometrics – active biometrics - Biometrics Vs traditional techniques – Benefits of biometrics - Operation of a biometric system– Key biometric processes: verification, identification and biometric matching – Performance measures in biometric systems: FAR, FRR, FTE rate, FTA rate and rate- Need for strong authentication – Protecting privacy and biometrics and policy – Biometric applications		
Module II	FINGERPRINT IDENTIFICATION TECHNOLOGY	9 Hours
Fingerprint Patterns, Fingerprint Features, Fingerprint Image, width between two ridges - Fingerprint Image Processing - Minutiae Determination - Fingerprint Matching: Fingerprint Classification, Matching policies.		
Module III	FACE RECOGNITION	9 Hours
Introduction, components, Facial Scan Technologies, Face Detection, Face Recognition, Representation and Classification, Kernel- based Methods and 3D Models, Learning the Face Spare, Facial Scan Strengths and Weaknesses, Methods for assessing progress in Face Recognition.		
Module IV	VOICE SCAN	9 Hours
Introduction, Components, Features and Models, Addition Method for managing Variability, Measuring Performance, Alternative Approaches, Voice Scan Strengths and Weaknesses, NIST Speaker Recognition Evaluation Program, Biometric System Integration		
Module V	FUSION IN BIOMETRICS	9 Hours
Introduction to Multibiometric - Information Fusion in Biometrics - Issues in Designing a Multi-biometric System - Sources of Multiple Evidence - Levels of Fusion in Biometrics - Sensor level, Feature level, Rank level, Decision level fusion - Score level Fusion. Examples – biopotential and gait based biometric systems.		

Total: 45 Hours

FURTHER READINGS: Germany Weighs Biometric Registration Options for Visa Applicants

Text Books:	
1.	James Wayman, Anil Jain, Davide Maltoni, Dario Maio, —Biometric Systems, Technology Design and Performance Evaluation, Springer, 2005.
2.	David D. Zhang, —Automated Biometrics: Technologies and Systems, Kluwer Academic Publishers, New Delhi, 2000.
3.	Arun A. Ross, Karthik Nandakumar, A.K.Jain, —Handbook of Multibiometrics, Springer, New Delhi, 2006.
References:	
1.	James Wayman, Anil Jain, Davide Maltoni, Dario Maio, —Biometric Systems, Technology Design and Performance Evaluation, Springer, 2005.
2.	David D. Zhang, —Automated Biometrics: Technologies and Systems, Kluwer Academic Publishers, New Delhi, 2000.
3.	Paul Reid, —Biometrics for Network Security, Pearson Education, 2004.
4.	Nalini K Ratha, Ruud Bolle, —Automatic fingerprint Recognition System, Springer, 2003.
5.	L C Jain, I Hayashi, S B Lee, U Halici, —Intelligent Biometric Techniques in Fingerprint and Face Recognition, CRC Press, 1999.

2303BM012	WEARABLE DEVICES (For B.E-BME)	L	T	P	C
		3	0	0	3
PRE REQUISITE:					
	Biosensors and measurements				
COURSE OBJECTIVES:					
	1.To understanding the technology behind wearable devices, including sensors, connectivity, and data processing.				
	2.To Understand how wearable devices collect, transmit, and analyze data, including issues related to data privacy and security.				
	3.To Analyze various applications of wearable devices across different sectors, such as healthcare, fitness, and industrial uses.				
COURSE OUTCOMES:					
On the successful completion of the course the students will be able to,					
CO1	Describe different sensors used in physiological system measurement				
CO2	Understand the bio signal acquisition methods and signal processing methods for human				

	systems
C03	Analyze the usage of optimized energy techniques for wearable Devices
C04	Evaluate the wireless Tele health technology for applications
C05	Executing the specified wearable systems for respective physiological systems.

COs Vs POs MAPPING:

CO s	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	1	-	-		1	-	1	2
CO2	3	2	1	1	1	-	-		1	-	1	2
CO3	3	2	1	1	1	-	-		1	-	1	2
CO4	3	2	1	1	1	-	-		1	-	1	2
CO5	3	2	1	1	1	-	-		1	-	1	2

COs Vs PSOs MAPPING:

CO s	PSO1	PSO2	PSO3
CO1	1		1
CO2	1		1
CO3	1		1
CO4	1		1
CO5	1		1

COURSE CONTENTS:

Module I	SENSORS	9 Hours
Need for wearable systems, Sensors for wearable systems-Inertia movement sensors, Respiration activity sensor, Inductive plethysmography, Impedance plethysmography, pneumography, Wearable ground reaction force sensor, GSR, Radiant thermal sensor, Wearable motion sensors, CMOS –Based Biosensors, E-Textiles, Bio compatibility.		
Module II	SIGNAL PROCESSING	9 Hours
Wearability issues -physical shape and placement of sensor, Technical challenges - sensor design, signal acquisition, Constraint on sampling frequency for reduced energy consumption, light weight signal processing, Rejection of irrelevant information, Datamining		
Module III	ENERGY HARVESTING FOR WEARABLE DEVICES	9 Hours
Solar cell, Vibration based Thermal based Human body as a heat source for power generation, Hybrid thermoelectric photovoltaic energy harvests, Thermopiles		
Module IV	WIRELESS HEALTH SYSTEMS	9 Hours
Need for wireless monitoring, Definition of Body area network and Healthcare, Technical Challenges- System security and reliability, BAN Architecture – Introduction, Wireless communication techniques.		
Module V	APPLICATIONS OF WEARABLE SYSTEMS	9 Hours
Medical Diagnostics, Medical Monitoring-Patients with chronic disease, Multi parameter monitoring, Neural recording, Gait analysis, Sports Medicine, Smart Fabrics.		

Total: 45 Hours

FURTHER READINGS: Flexible electronics

Text books:

-
1. Annalisa Bonfiglio, Danilo De Rossi, "Wearable Monitoring Systems", Springer, 2011
-
2. Sandeep K.S. Gupta, Tridib Mukherjee, Krishna Kumar Venkata Subramanian, "Body Area Networks Safety, Security, and Sustainability," Cambridge University Press, 2013.

References:

1. Hang, Yuan-Ting, "wearable medical sensors and systems", Springer-2013
2. Guang-Zhong Yang (Ed.), "Body Sensor Networks, "Springer, 2006

3.Mehmet R. Yuce, Jamil Y. Khan,"Wireless Body Area Networks Technology, Implementation and Applications " , Pan Stanford Publishing Pvt. Ltd, Singapore,2012
4.Andreas Lymberis, Danilo de Rossi ,'Wearable eHealth systems for Personalised Health Management - State of the art and future challenges ' IOS press, The Netherlands,2004

OPEN ELECTIVE

2303BM013	OCCUPATIONAL SAFETY AND HEALTH IN PUBLIC HEALTH EMERGENCIES	L 3	T 0	P 0	C 3
PRE REQUISITE:					
	Clinical engineering				
	Hospital training and management				
COURSE OBJECTIVES:					
	1.To know about Occupational safety and health (OSH)				
	2.Discuss about risks faced by emergency responders during disease outbreaks and other emergencies				
	3.To create awareness on necessary strategies for managing OSH in emergency situations				
COURSE OUTCOMES:					
On the successful completion of the course the students will be able to,					
CO1	Practice the occupational safety measures by the scientific knowledge to overcome the risks faced by emergency responders				
CO2	Apply appropriate strategies and tools in Occupational safety and healthcare				
CO3	Analyze common risks for safety and health in emergencies				

CO4	Adapt appropriate occupational safety practices in chemical accidents												
CO5	Guide Occupational safety measures in radiation incidents												
COs Vs POs MAPPING:													
	CO No	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
	1	3	2	2	1				1			1	2
	2	3	2	2	2				1			1	2
	3	3	2	2	2				1			1	2
	4	3	2	2	2				1			1	2
	5	3	2	2	2				1			1	2
COs Vs PSOs MAPPING													
	COs	PSO1	PSO2	PSO3									
	CO1		2										
	CO2		2										
	CO3		2										
	CO4		2										
	CO5		2										
COURSE CONTENTS:													
Module I	MANAGEMENT ASPECTS											9 Hours	
Management system approach to occupational safety and health hazards and risks – rights, duties and responsibilities of employers and workers during outbreaks and emergencies –Emergency responders health monitoring and surveillance													
Module II	STRATEGIES AND TOOLS											9 Hours	
International Health Regulations, 2005 – Incident command system for managing outbreaks and emergencies–Occupational safety and health controls–Strategies for infection prevention and control													
Module III	COMMON RISKS FOR SAFETY AND HEALTH IN EMERGENCIES											9 Hours	
Vector-borne diseases, water and food-borne diseases,Vaccine-preventable diseases–Heatstress- Slips, trips and falls - Road traffic injuries – Ergonomic hazards - Violence – Psychological stress during outbreaks and injuries													
Module IV	OCCUPATIONAL SAFETY AND HEALTH IN CHEMICAL INCIDENTS											9 Hours	
Emergencies caused by chemical incidents – Occupational safety and health hazards and risks of chemicals–Personal Protective Equipment –Decontamination of emergency response personal – Medical surveillance of emergency responders													
Module V	OCCUPATIONAL SAFETY AND HEALTH IN RADIATION INCIDENTS											9 Hours	
Sources and scenarios of radiation incidents–Guidance for protection of emergency responders- Occupational health surveillance of persons occupationally exposed to radiation in emergencies													
Total: 45 Hours													
FURTHER READINGS: Emergency medicine													
References:													
1.Emergency responder health monitoring and surveillance. National Response Team technical assistance document. Atlanta (GA): National Institute for Occupational Safety and Health; 2012.													
2.Emergency response framework(ERF).Geneva: World Health Organization;2013													
3.Guidelines on occupational safety and health management systems, second edition. Geneva:													

International Labour Organization; 2009.
4.OSH management system: a tool for continual improvement. Geneva: International Labour Organization; 2011
5.OECD Environmental Outlook to 2050: the consequences of inaction. Paris: Organization for Economic Co-operation and Development; 2012.